

Media Bias ISM: Master Methods & Results Table

Shared notation. i firm, t day/week. α_i firm FE, δ_t time (day/month) FE. $\text{Post}_t = \mathbb{I}[t \geq 2020-01-01]$. $s_a = p_a^+ - p_a^-$ (NewsMTSC article stance); $\text{bias}_{it} = |\mathcal{A}_{it}|^{-1} \sum_a s_a$; Δbias_{it} first difference (used in VAR); $\text{vol}_{it} = |\mathcal{A}_{it}|$ (firm-day articles); $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$ (weekly: summed). Inference: HC1 (panel regressions); HAC/Newey–West (VAR).

Keys. Verdict: **sig.** ($p < 0.05$); **marg.** ($0.05 \leq p < 0.10$); **n.s.** ($p \geq 0.10$); n/a descriptive/no test. Fit: within- R^2 for regressions; VAR uses a Granger F -test, so R^2 is **n/a**. [\[check\]](#) construction/sanity check (shown first in each RQ). [\[legacy\]](#) superseded sample, retained for reference. Corpora (RQ4): **All-art**: all scored articles; **Rel.**: ensemble relevance-filtered (121,177 / 695,731).

ID	Method / spec	Equation	Sample	Headline result	Fit (R^2)	Verd.
RQ1: Did the tone of media coverage of S&P 500 firms shift after Jan 2020? Outcome: daily firm-day stance bias_{it} . Construction: $s_a = p^+ - p^-$ (NewsMTSC), high-confidence outputs, averaged per firm-day.						
RQ1.C1	[check] Pre-period mean equality	Tolerance rule on firm pre-period stance means	26 firms; $\tau=0$	Means <i>unequal</i> (range 0.66, std 0.137) $\Rightarrow$ SCDiD fallback considered for clean identification.	n/a	n/a
RQ1.C2	[check] Joint F -test on day FE	$H_0 : \delta_1 = \dots = \delta_T = 0$	df (4,017, 59,181)	$F=1.135$, $p=1.1 \times 10^{-8}$; $F_{\text{crit}}=1.038$. Day FE jointly significant $\Rightarrow$ justifies two-way FE (RQ1.1).	n/a	sig.
RQ1.C3	[check] MLR / Gauss–Markov diagnostics	Breusch–Pagan; Jarque–Bera; VIF; Durbin–Watson on RQ1.1 residuals	$n=63,200$	BP $p < 10^{-42}$ (heterosk.); JB $p \approx 0$ (fat tails, kurt 4.25); VIF ≈ 1.0 ; DW = 1.673. $\Rightarrow$ HC1 justified.	n/a	n/a
RQ1.1	Two-way FE (firm + day)	$\text{bias}_{it} = \alpha_i + \delta_t + \beta \text{Post}_t + \gamma \text{vol}_{it} + \varepsilon_{it}$	$n=63,200$; 26 firms; 4,018 days; $\tau=0$	$\beta_{\text{Post}} = +0.0141$; HC1 SE 0.0136, $p=0.30$; day-clustered SE 0.0134, $p=0.29$. vol -0.00097^{***} . Post insignificant once daily common shocks absorbed.	0.0017	n.s.
RQ1.1b	Two-way FE, log-volume control	$\text{bias}_{it} = \alpha_i + \delta_t + \beta \text{Post}_t + \gamma \ln(1+\text{vol}_{it}) + \varepsilon_{it}$	$n=63,200$; $G=4,018$	$\beta_{\text{Post}} = +0.0146$ (SE 0.0134), $p=0.28$; ln-vol -0.0161^{***} . Robustness to volume scaling.	0.0014	n.s.
RQ1.4	Monthly event study on stance	$\text{bias}_{it} = \alpha_i + \sum_{k \neq -1} \beta_k \mathbb{I}[t \in \text{mo}_k] + \gamma \text{vol}_{it} + \varepsilon_{it}$	$n=23,439$; $k \in [-24, +24]$; ref $k=-1$	Pre-trend ≈ 0 . $\beta_0 = +0.031$ ($p=0.09$); $\beta_{+12} = +0.064$ ($p < 10^{-3}$); persists $+0.05$ to $+0.10$ to $k=+24$.	n/a	sig.
RQ1.L	[legacy] Confidence-threshold sweep, one-way firm FE	$\text{bias}_{it} = \alpha_i + \beta \text{Post}_t + \gamma_1 \text{vol}_{it} + \gamma_2 \text{vix}_t + \varepsilon_{it}$ (no day FE)	$\tau=0/0.6/0.9$; $n=49,136$ / 46,681 / 31,575; 22 firms	$\beta_{\text{Post}} = +0.0208$ / $+0.0232$ / $+0.0258$ (per τ); all $p < 10^{-6}$. Significant on the earlier one-way-FE / 22-firm sample; superseded by RQ1.1 (effect vanishes once day FE added).	0.0018–0.0026	sig.
RQ2: Did firm-level returns shift after Jan 2020 beyond market co-movement? Outcome: daily return return_{it} . Construction: $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$ from <i>stock.prices</i> close; market controls S&P 500 return + VIX.						
RQ2.C1	[check] Pre-period mean equality	Tolerance rule on firm pre-period return means	24 firms	Means <i>equal</i> (range 0.0019, std 0.0004) $\Rightarrow$ parallel-trends plausible; standard DiD valid, SCDiD not triggered.	n/a	n/a
RQ2.1	Baseline firm FE on returns	$\text{return}_{it} = \alpha_i + \beta \text{Post}_t + \gamma_1 \text{sp500_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}$	$n=62,667$; 26 firms; 2015-12 to 2025-11	$\beta_{\text{Post}} = -0.000344$ (SE 0.000172), $p=0.045$ (≈ -3.4 bps/day); S&P $\beta=1.128^{***}$; VIX n.s.	n/r	marg.
RQ2.2	Two-way FE (firm + month)	$\text{return}_{it} = \alpha_i + \delta_t + \beta \text{Post}_t + \gamma_1 \text{sp500_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}$	$n=62,667$	$\beta_{\text{Post}} = -0.0018$ (SE 0.0142), $p=0.90$; not significant once month FE included. S&P ret absorbed by month FE ($\beta \approx 0$).	0.0000 ¹	n.s.

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Master methods table (continued)

ID	Method / spec	Equation	Sample	Headline result	Fit (R^2)	Verd.
RQ2.3	[legacy] Monthly event study on returns	$\text{return}_{it} = \alpha_i + \sum_{k \neq -1} \beta_k \mathbb{I}[t \in \text{mo}_k] + \gamma_1 \text{sp500_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}$	$n=62,667$; ref Dec 2019	Pre-trend flat. β_0 (Jan 2020) = -0.00145 , $p=0.044$; post bins mostly n.s. (<i>no result file in results/rq2/; verify before citing.</i>)	n/a	marg.
RQ3: Does media bias predict returns at all? Weekly VAR($p=3$) + Granger causality (HAC SE), 15 firms. Both corpora reported. <i>Construction: ensemble relevance filter (both corpora); weekly ISO aggregation + gap imputation; Δbias (bias $I(1) \rightarrow$ differenced); weekly log returns winsorized $\pm 5SD/\text{firm}$ (excl. AT&T); Helmert FE removal.</i>						
RQ3.C1	[check] Stationarity (ADF + KPSS)	ADF H_0 : unit root; KPSS H_0 : stationary	16 firms, weekly	Returns $I(0)$ all 16. Bias levels $I(1)$ /“conflicting” 14/16 $\Rightarrow$ VAR uses Δ bias.	n/a	n/a
RQ3.C2	[check] VAR lag selection & residual diagnostics	BIC lag order; stability roots; Portmanteau; Jarque–Bera; ARCH	15 firms	Modal BIC lag $p=3$; all stable. Portmanteau fails 11/15; JB fails all; ARCH 10/15 $\Rightarrow$ HAC/Newey–West SE throughout.	n/a	n/a
RQ3.1	Panel VAR Granger: Bias $\rightarrow$ Returns	Pooled VAR on $[\Delta \text{bias}_{it}, \text{return}_{it}]$; Helmert FE; +VIX +S&P ret + article count	pooled firm-weeks; 15 firms	All-art: $p=0.976$ / 0.245 / 0.617 (full/pre/post); $N=7,493$. Rel.: $p=0.114$ / 0.009*** / 0.672; $N=6,976$. Significant pre-break only under filtering.	n/a (F -test)	sig.
RQ3.2	Panel VAR Granger: Returns $\rightarrow$ Bias	(reverse direction of RQ3.1)	15 firms	All-art: $p=0.200$ / 0.801 / 0.187. Rel.: $p=0.706$ / 0.359 / 0.131. Reverse channel null in both corpora.	n/a (F -test)	n.s.
RQ3.3	Firm Granger: Bias $\rightarrow$ Returns (sig. firms)	Per-firm bivariate VAR, HAC SE; full / pre / post-break	≥ 100 wks/firm	All-art: WFC full 0.023**, post 0.013**; MS full 0.058*; MCD post 0.077*. Rel.: SBUX full 0.024**, pre 0.014**; MS pre 0.001*** ; no firm sig. post-break.	n/a (F -test)	sig.
RQ3.4	Firm Granger: Returns $\rightarrow$ Bias (sig. firms)	Per-firm bivariate VAR, HAC SE	≥ 100 wks/firm	All-art: GS full 0.012** / post 0.007***; MS post 0.016**; BA post 0.014**; MCD pre 0.027**. Rel.: GS full 0.032** / pre 0.010***; GM full 0.079*. Price $\rightarrow$ coverage for financials.	n/a (F -test)	sig.
RQ3.5	Firm Granger: remaining firms	Per-firm bivariate VAR, HAC SE	both corpora	ABNB, AMZN, BAC, INTC, MSFT, T, UBER, V: no significant Granger in either direction. NFLX dropped (coverage); WMT excluded (data bug).	n/a (F -test)	n.s.
RQ4: Did the bias$\rightarrow$return link strengthen post-2020? Interpretation of the pre/post Granger split (no new estimation; draws on RQ3). <i>Split: firm-specific Bai–Perron single break in the bias series (see RQ4.C1), not a common Jan 2020 date.</i>						
RQ4.C1	[check] Structural breaks (Bai–Perron)	Single largest mean shift per firm in the bias series	16 firms	Breaks span 2018–2024 (only V $\approx$ Sep 2020). “Pre-break” $\neq$ “pre-COVID” $\Rightarrow$ calendar reading needs a fixed-date robustness check.	n/a	n/a
RQ4.1	Pre vs. post-break Granger shift (from RQ3)	Compare Bias $\rightarrow$ Returns Granger p across the firm-specific break	panel + firm; both corpora	All-art: panel pre $p=0.245 \rightarrow$ post 0.617 (no change); firm WFC pre 0.345 $\rightarrow$ post 0.013**. Rel.: panel pre 0.009*** $\rightarrow$ post 0.672; firms MS/SBUX significant pre-break, none post. Effect is pre-break , not strengthening after 2020.	n/a	marg.

Cross-cutting reading. Static pre/post dummies are significant on the one-way firm-FE design (RQ1.L) but weaken or vanish once daily common shocks are absorbed (RQ1.1, RQ2.2); a single Post dummy averages five heterogeneous post-shock years into one coefficient. The monthly event study (RQ1.4) shows the stance shift is gradual, rising to $\beta_{+12}=+0.064$ by year 1, while returns show a one-month hit at the event date. The weekly VAR (RQ3) is run on two corpora. On all scored articles, media bias does not predict returns at the panel level ($p=0.976$). On relevance-filtered articles the effect is significant in the pre-break period only (panel $p=0.009$; firms MS, SBUX) and null post-break ($p=0.672$), which is the reverse of a post-2020 strengthening (RQ4). Because the structural breaks are firm-specific and span

¹RQ2.2 within- $R^2 \approx 0$ because month fixed effects absorb the S&P 500 return (the dominant regressor); the model’s explanatory power is loaded onto the FE, not the within variation.

2018 to 2024 (RQ4.C1), “pre-break” is not the same as “pre-COVID,” so the calendar reading needs a fixed-date robustness check. Reverse causality (returns→bias) is significant only for financials (GS, MS), consistent with price moves preceding coverage.